

exploratory1.ipynb

15 cells · python

INITIALIZATION & ENVIRONMENT SETUP

IN [45]:

```
# Core numerical, data manipulation and visualization frameworks

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
sns.set_style('darkgrid')
```

IN [46]:

```
# Load transformed and feature-engineered dataset
data = pd.read_csv('../data/bank_churn_data_cleaned.csv',
                    dtype={'CustomerId': str})
```

```
df = pd.DataFrame(data)
```

```
df.head()
```

OUTPUT

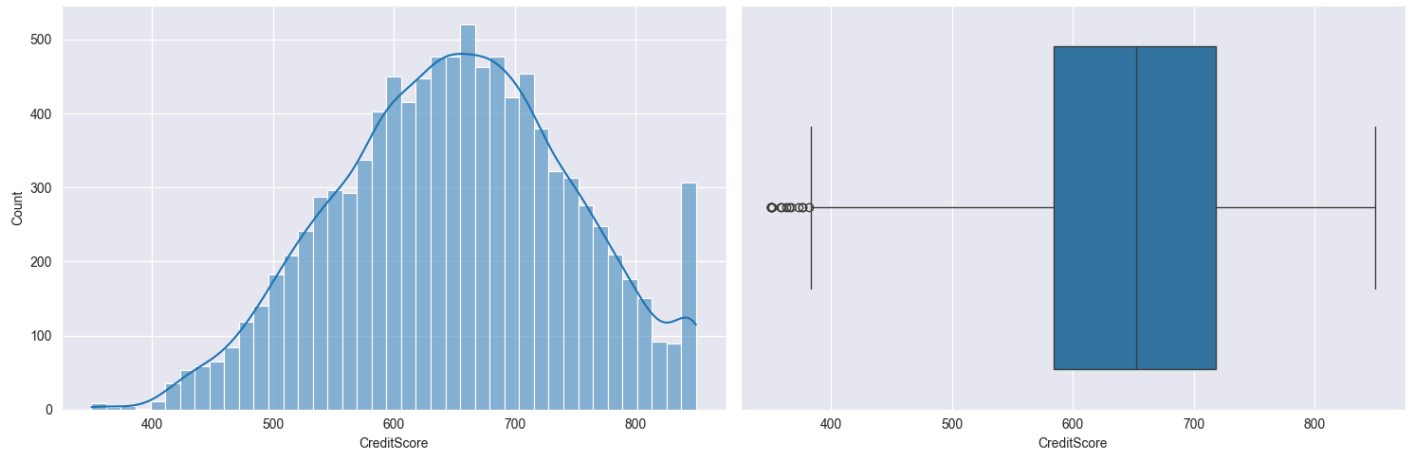
	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOff
0	15634602	Hargrave	619	France	Female	42	2	0.00	1
1	15647311	Hill	608	Spain	Female	41	1	83807.86	1
2	15619304	Onio	502	France	Female	42	8	159660.80	3
3	15701354	Boni	699	France	Female	39	1	0.00	2
4	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1

UNIVARIATE ANALYSIS

IN [47]:

```
# Output CreditScore distribution dynamics and structural outlier analysis
fig, axes = plt.subplots(1,2, figsize= (15,5))
sns.histplot(data=df, ax=axes[0], x='CreditScore', kde=True)
sns.boxplot(data=df, x='CreditScore', ax= axes[1])
plt.tight_layout()
plt.show()
```

OUTPUT

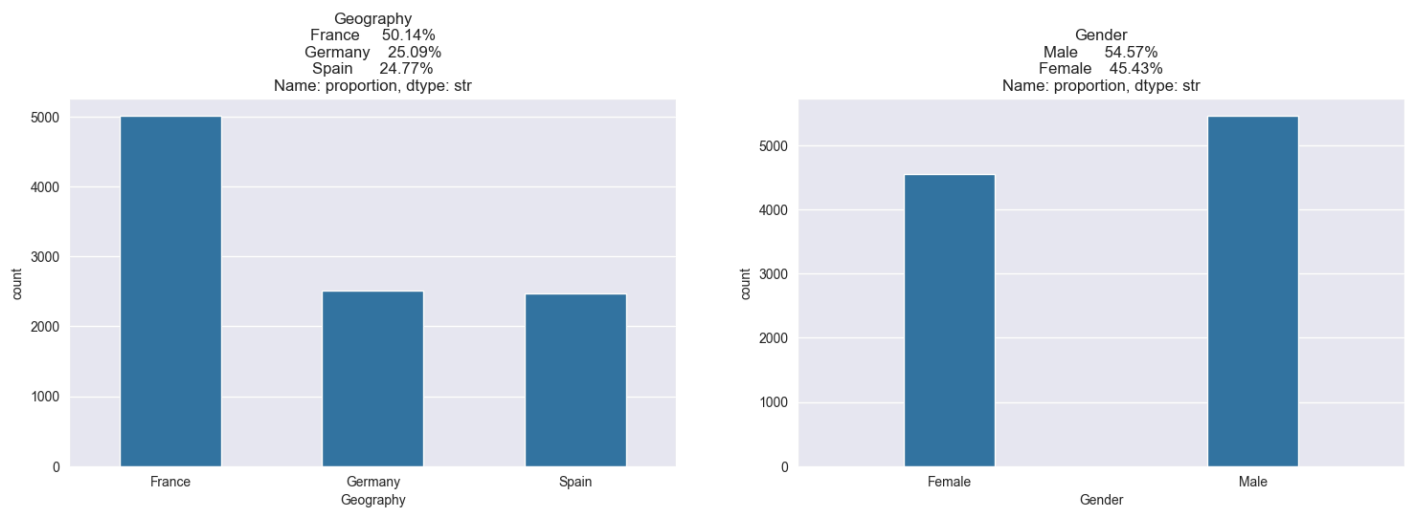


IN [48]:

```
# Output geographic concentration patterns and gender demographic splits
fig, axes = plt.subplots(1,2, figsize= (18, 5))
sns.countplot(data=df, x='Geography', ax=axes[0], width=0.5, order=['France', 'Germany',
'Spain'])
axes[0].set_title(f'{{(df['Geography'].value_counts(normalize=True)*100).round(2).astype(str)
+ '%'}}')
sns.countplot(data=df, x='Gender', ax=axes[1], width=0.3)
axes[1].set_title(f'{{(df['Gender'].value_counts(normalize=True)*100).round(2).astype(str)
+ '%'}}')

plt.show()
```

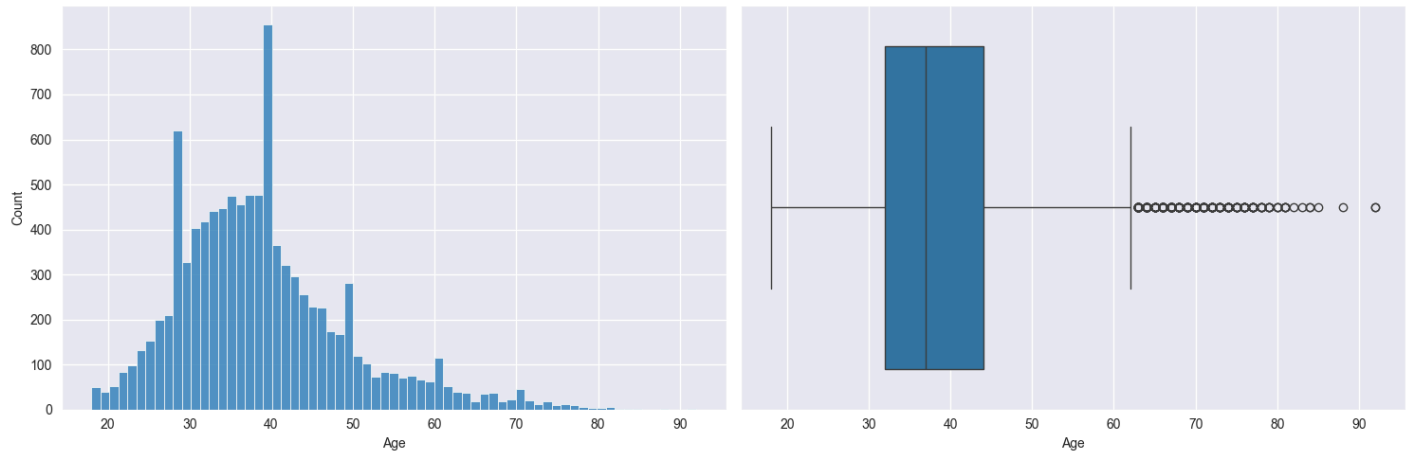
OUTPUT



IN [49]:

```
# Customers' age distribution
fig, axes = plt.subplots(1,2, figsize= (15,5))
sns.histplot(data=df, ax=axes[0], x='Age')
sns.boxplot(data=df, x='Age', ax= axes[1])
plt.tight_layout()
plt.show()
```

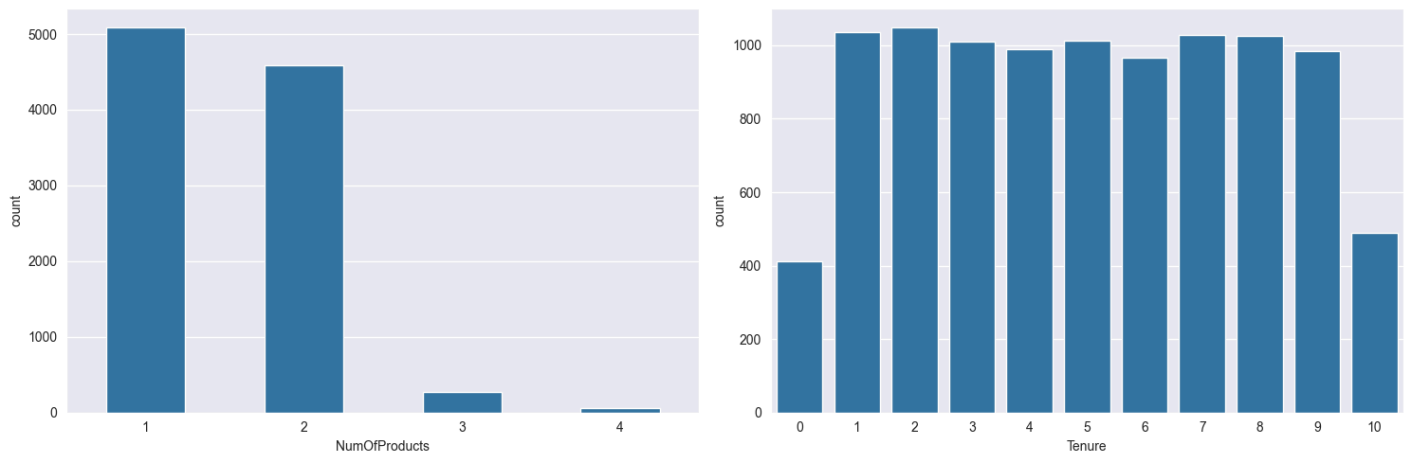
OUTPUT



IN [50]:

```
# Customers' product density and tenure footprints
fig, axes = plt.subplots(1,2, figsize= (15,5))
sns.countplot(data=df, ax=axes[0], x='NumOfProducts', width=0.5)
sns.countplot(data=df, x='Tenure', ax= axes[1])
plt.tight_layout()
plt.show()
```

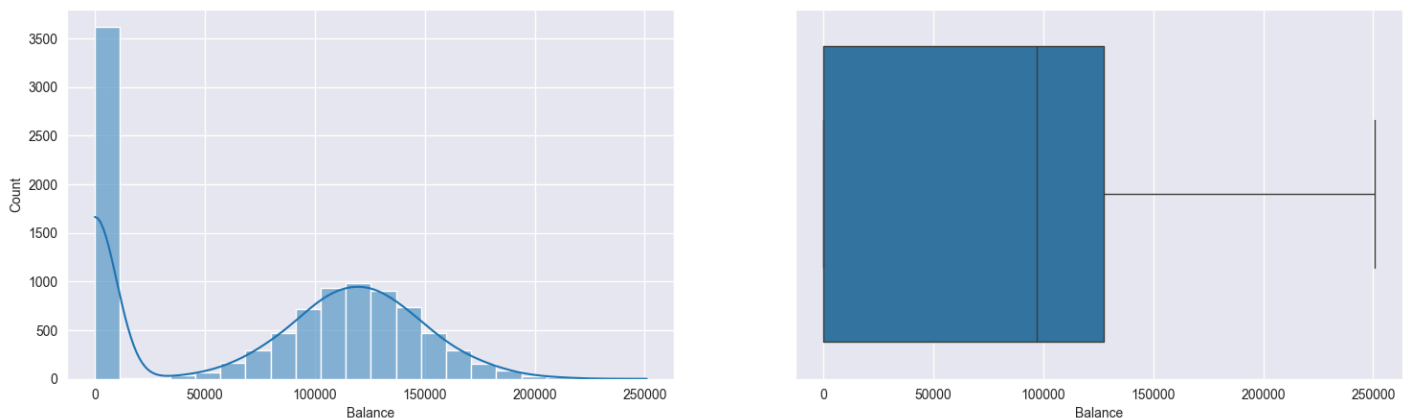
OUTPUT



IN [51]:

```
# Output balance concentration
fig, axes = plt.subplots(1,2, figsize=(18, 5))
sns.histplot(data=df, x='Balance', ax=axes[0], kde=True)
sns.boxplot(data=df, x='Balance', ax=axes[1])
plt.show()
```

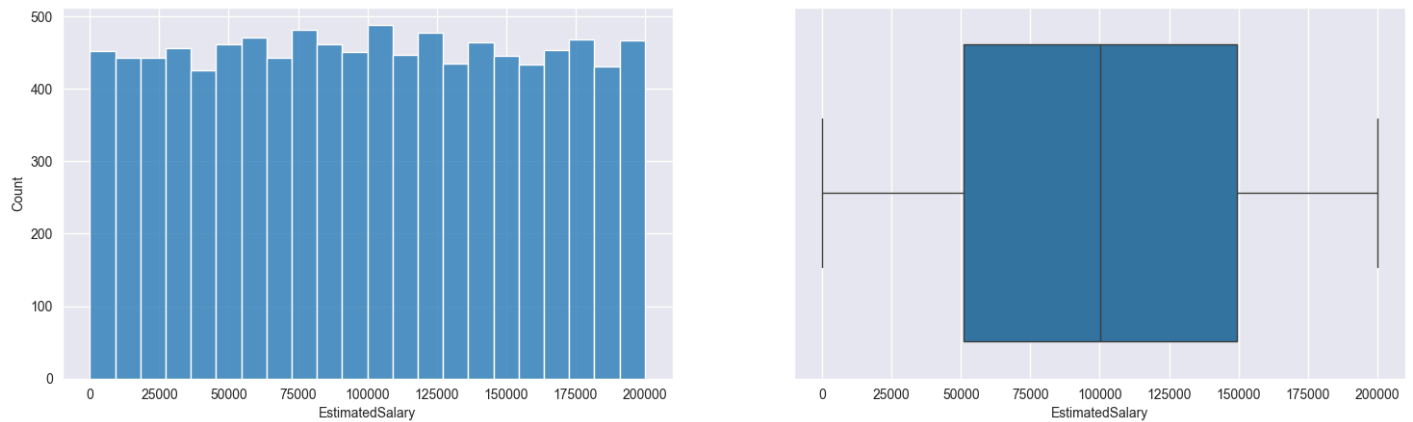
OUTPUT



IN [52]:

```
# Output estimated salary and economic earning distribution
fig, axes = plt.subplots(1,2, figsize=(18, 5))
sns.histplot(data=df, x='EstimatedSalary', ax=axes[0])
sns.boxplot(data=df, x='EstimatedSalary', ax=axes[1])
plt.show()
```

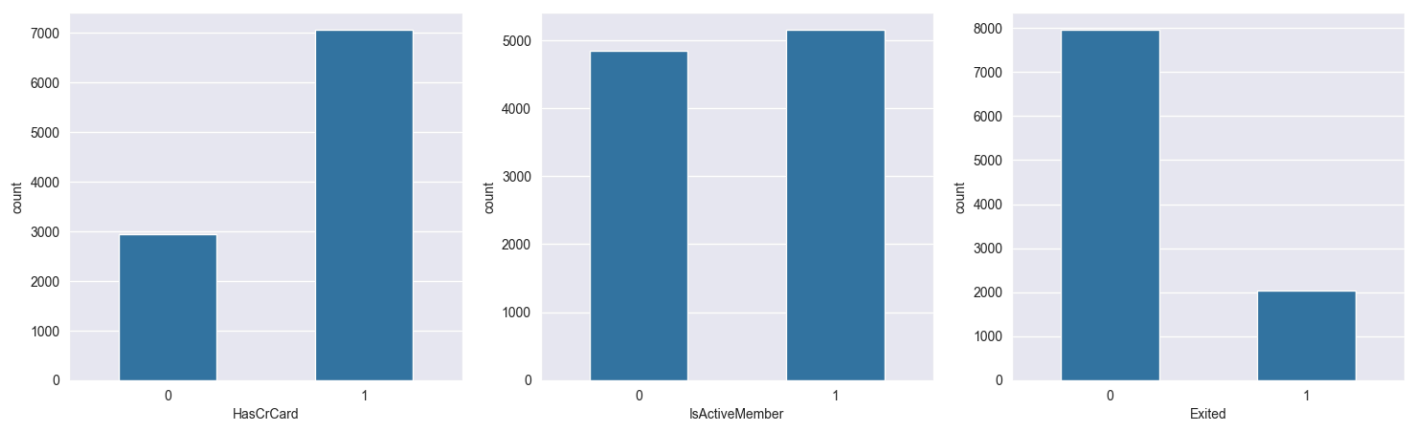
OUTPUT



IN [53]:

```
# Output financial tool coverage, platform interaction metrics, and baseline attrition
metrics
fig, axes = plt.subplots(1,3, figsize=(18, 5))
sns.countplot(data=df, x='HasCrCard', ax=axes[0], width=0.5)
sns.countplot(data=df, x='IsActiveMember', ax=axes[1], width=0.5)
sns.countplot(data=df, x='Exited', ax=axes[2], width=0.5)
plt.show()
```

OUTPUT



IN [54]:

```
print((df['IsActiveMember'].value_counts(normalize=True)*100).round(2).astype(str) + '%')
print('='*175)

print((df['Exited'].value_counts(normalize=True)*100).round(2).astype(str) + '%')
print('='*175)

print((df['HasCrCard'].value_counts(normalize=True)*100).round(2).astype(str) + '%')
```

OUTPUT

IsActiveMember

1 51.51%

0 48.49%

Name: proportion, dtype: str

=====

Exited

0 79.63%

1 20.37%

Name: proportion, dtype: str

=====

HasCrCard

1 70.55%

0 29.45%

Name: proportion, dtype: str

◀  ▶

VISUALIZATION

IN [55]:

```
# DASHBOARD A: THE BANK'S DEMOGRAPHIC PROFILE

fig, axes = plt.subplots(2, 2, figsize=(15, 9))
fig.suptitle('Executive Dashboard A: Customer Demographic', fontsize=20,
fontweight='bold', y=0.95)

# Axis (0,0): Age Distribution
sns.histplot(data=df, x='Age', kde=True, ax=axes[0, 0], color='brown', alpha=0.6)
axes[0, 0].set_title('Age Distribution Profile', fontsize=14, fontweight='semibold')
axes[0, 0].set_xlabel('Age')

# Axis (0,1): Gender Representation
sns.countplot(data=df, x='Gender', ax=axes[0, 1], palette='Oranges_r', width=0.4,
hue='Gender')
axes[0, 1].set_title('Gender Structural Volume Split', fontsize=14, fontweight='semibold')
axes[0, 1].set_xlabel('Gender')

# Axis (1,0): Geographical Dispersion
sns.countplot(data=df, x='Geography', ax=axes[1, 0], palette='OrRd_r', width=0.5, order=
['France', 'Germany', 'Spain'], hue='Geography')
axes[1, 0].set_title('Geographic Dispersion Split', fontsize=14, fontweight='semibold')
axes[1, 0].set_xlabel('Country Location')

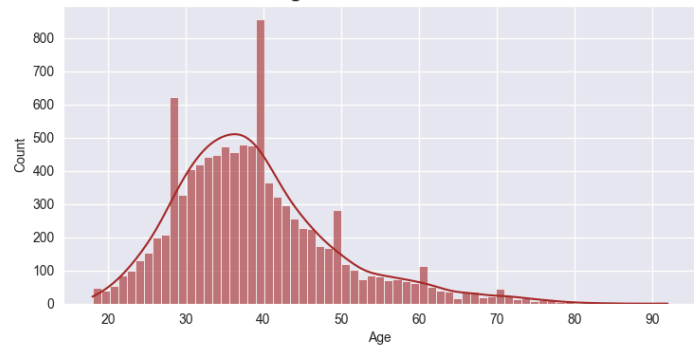
# Axis (1,1): Brand Relationship Longevity
sns.countplot(data=df, x='Tenure', ax=axes[1, 1], color='brown', alpha=0.7)
axes[1, 1].set_title('Customer Account Tenure (Years)', fontsize=14,
fontweight='semibold')
axes[1, 1].set_xlabel('Tenure Cohort')

plt.tight_layout(rect=[0, 0, 1, 0.94])
plt.savefig('../visuals/customer_demographic_profile.png', dpi=300, bbox_inches='tight')
plt.show()
```

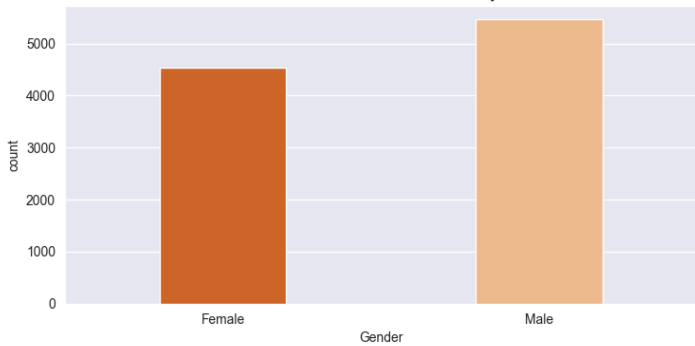
OUTPUT

Executive Dashboard A: Customer Demographic

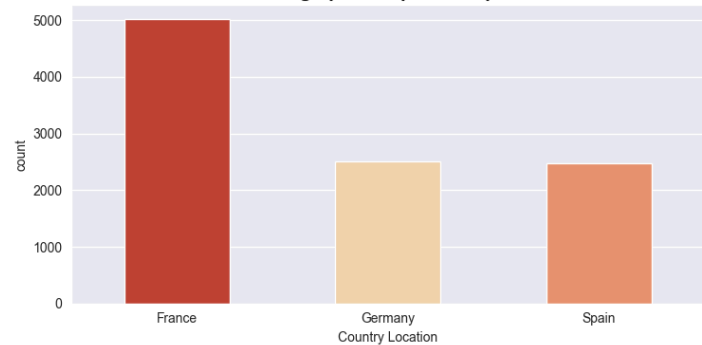
Age Distribution Profile



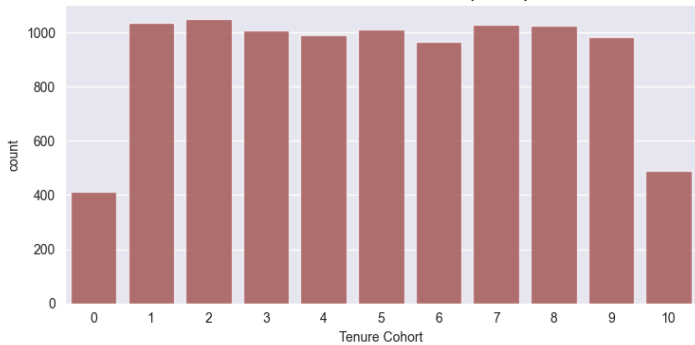
Gender Structural Volume Split



Geographic Dispersion Split



Customer Account Tenure (Years)



IN [56]:

```
# DASHBOARD B: THE BANK'S FINANCIAL PROFILE

fig, axes = plt.subplots(2, 2, figsize=(15, 9))
fig.suptitle('Executive Dashboard B: Financial Account Footprint', fontsize=20,
fontweight='bold', y=0.95)

# Axis (0,0): Consumer Credit Health
sns.histplot(data=df, x='CreditScore', kde=True, ax=axes[0, 0], color='brown')
axes[0, 0].set_title('Consumer Credit Score Distribution Spectrum', fontsize=14,
fontweight='semibold')
axes[0, 0].set_xlabel('Credit Score')

# Axis (0,1): Balance Concentrations
sns.histplot(data=df, x='Balance', kde=True, ax=axes[0, 1], color='red', alpha=0.5)
axes[0, 1].set_title('Account Balance Asset Concentrations', fontsize=14,
fontweight='semibold')
axes[0, 1].set_xlabel('Balance Amount')

# Axis (1,0): Salary Uniformity Assessment
sns.histplot(data=df, x='EstimatedSalary', ax=axes[1, 0], color='Maroon', bins=30,
alpha=0.6)
axes[1, 0].set_title('Estimated Earning Profiles', fontsize=14, fontweight='semibold')
axes[1, 0].set_xlabel('Estimated Earning')

# Axis (1,1): Product Distribution Check
sns.countplot(data=df, x='NumOfProducts', ax=axes[1, 1], palette='Reds_r', width=0.5,
hue='NumOfProducts', alpha=0.6)
axes[1, 1].set_title('Distribution of Product Ownership', fontsize=14,
fontweight='semibold')
axes[1, 1].set_xlabel('Banking Products')

plt.tight_layout(rect=[0, 0, 1, 0.94])
plt.savefig('../visuals/financial_account_footprint.png', dpi=300, bbox_inches='tight')
plt.show()
```

OUTPUT

Executive Dashboard B: Financial Account Footprint

